MULTIPLE LINEAR REGRESSION

with Multiple Variables

🎯 ML Course — Lecture 4

📊 **Assessing the Model**

R2, MSE, RMSE, MAE

🏠 **Polynomial Regression**

Google colab Examples

Prerequisites: Simple Linear Regression — Session 4

Assessing the Model

Assessing the Model

The most important performance metric from a data science perspective is *root mean squared error*, or *RMSE*. RMSE is the square root of the average squared error in the predicted $\hat{y}_i$ values:

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n} (y_i - \hat{y}_i)^2}{}}$$

`scikit-learn` provides a number of metrics for regression and classification. Here, we use `mean_squared_error` to get RMSE and `r2_score` for the coefficient of determination:

```
fitted = house_lm.predict(house[predictors])
RMSE = np.sqrt(mean_squared_error(house[outcome], fitted))
r2 = r2_score(house[outcome], fitted)
print(f'RMSE: {RMSE:.0f}')
print(f'r2: {r2:.4f}')
```

Multiple Linear Regression – R2 (R^2)

- Another useful metric is the coefficient of determination, also called the **R-squared statistic** or R^2.
- Measures **the proportion of variation** in the data that is accounted for in the model.
- It is useful mainly in explanatory uses of regression where you want to **assess how well the model fits the data**.

$$R^2 = 1 - \frac{\Sigma_{i=1}^{n}(y_i - \hat{y}_i)^2}{\Sigma_{i=1}^{n}(y_i - \bar{y})^2}$$

- Its value ranges from 0 to 1:
  - 0 means no relationship
  - 1 means totally related.

👉 However, it can be negative if the model performs very badly.

Multiple Linear Regression – R2 (R^2)

👉 **R^2 tells how much of the output is explained by the input using the model**

👉 R^2 does NOT tell if model is correct, only how well it fits data.

👉 High R^2 ≠ always good model (overfitting possible)

Multiple Linear Regression – R2 (R^2)

```
# R2
# How well does my training data fit in a polynomial regression?

import numpy
from sklearn.metrics import r2_score
numpy.random.seed(2)

x = numpy.random.normal(3, 1, 100)
y = numpy.random.normal(150, 40, 100) / x

train_x = x[:80]
train_y = y[:80]

test_x = x[80:]
test_y = y[80:]

mymodel = numpy.poly1d(numpy.polyfit(train_x, train_y, 4))

r2 = r2_score(train_y, mymodel(train_x))

print(r2)
```

R = 0.799 shows that there is a relationship.

Regression - Performance Metrics

Metric	Full Form	Formula	Range	Interpretation	Sensitivity to Outliers
MSE	Mean Squared Error	$\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$	$[0, \infty)$	Measures average squared error. Penalizes large errors. Lower is better.	High
RMSE	Root Mean Squared Error	$\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$	$[0, \infty)$	Square root of MSE, in same units as the target variable. Lower is better.	High
R^2	Coefficient of Determination	$1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \hat{y})^2}$	$(-\infty, 1]$	Indicates proportion of variance explained by the model. Closer to 1 is better.	Moderate
MAE	Mean Absolute Error	$\frac{1}{n}\sum_{i=1}^{n}\lvert y_i - \hat{y}_i \rvert$	$[0, \infty)$	Measures average absolute difference between actual and predicted values. Lower is better.	Low

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